Efficacy of a Gaussian Mixture Model based Speaker Recognition in Machine Learning Framework

Adhiraj Banerjee

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1 Introduction

Speaker Recognition (SR) has found numerous applications such as voice biometric, forensics, traitor finding, voice-mail and tele-banking. Conceptually, SR determines the presence of an active speaker among a set of registered speakers. In view of the applicability of Machine Learning framework to a wide variety of classification problems, in the present work emhasis is laid on the popular Gaussian Mixture Model (GMM) based ML (GMM-ML) approach as a potential solution for SR. Voice samples of ten different speakers are recorded in a python enviroment and the performance of GMM-ML for SR is analyzed. It is observed that GMM-ML based SR is highly accurate in distinguishing the actual speaker from the 10 registered speakers.

2 "Machine Learning" for Speaker Recognition

Machine Learning (ML) is a mathematical framework which helps systems to learn from trained model and predict the class into which the test data may be classified to. The present work studies the efficacy of a ML algorithm for Speaker Recognition. Going by principles of ML, a model for Speaker Recognition is trained such that the model identifies unique patterns (referred as features) from a set of different voice samples (of registered speakers) and distinguish the actual speaker voice sample from the rest. In the present work, a familiar ML algorithm, Gaussian Mixture Model (GMM) based classification, is chosen and its efficacy in accurate speaker recognition is studied.

2.1 Gaussian Mixture Model based SR

It is known that the Gaussian Mixture Model (GMM) is a powerful model used to solve clustering problems. It features a probabilistic approach by which it approximates the probability distribution of a MXK-order Data($\boldsymbol{M}$ is the number

of $\boldsymbol{K}$-dimensional data vectors) belonging to a class $\boldsymbol{\lambda}$ as a linear combination of $\boldsymbol{N}$ Gaussian Distributions or Clusters with mean vector $\boldsymbol{\mu}$=$\{\mu_i\}_{i=1,2,...,N}$ (i.e. order $KX1$), variance-covariance matrix $\boldsymbol{\Sigma}$ (i.e. order KXK) and $\boldsymbol{\omega}$=$\{\omega_i\}_{i=1,2,...,N}$ (i.e. order $KX1$) as the weights in the linear combination. In *Speaker Recognition*, features of speaker's voice sample are extracted into a feature vector and used as data points for processing to identify the registered speaker. Since the feature vector is *multi-dimensional*, the probability density function (PDF) of particular class of speaker (λ) can be expressed as a linear combination of N multivariate Gaussian PDFs as:

$$P(\boldsymbol{X}|\lambda) = \sum_{i=1}^{N} \omega_i \cdot P(\boldsymbol{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \tag{1}$$

and,

$$P(\boldsymbol{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \prod_{n=1}^{M} P(\boldsymbol{x}_n|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \tag{2}$$

where, M denotes the number of Gaussian random variables used for N clusters formed from the input vector $\boldsymbol{X}$; λ, as mentioned earlier, is the class to which a speaker is associated with; $\boldsymbol{X}$ represents the N dimensional training data; and $P(\boldsymbol{x}_n|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ is the multivariate Gaussian PDF given by,

$$P(\boldsymbol{x}_n|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \left((2\pi)^N \cdot |\boldsymbol{\Sigma}_i|\right)^{-0.5} \cdot exp\left(-0.5(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right). \tag{3}$$

In the formulation in Equation 1, the probability, also referred as the likelihood, computed from the input data points in vector X is assigned to a particular class corresponding to the speaker's GMM. Herein, the mean vectors ($\boldsymbol{\mu}_i$), variance-covariance matrices ($\boldsymbol{\Sigma}_i$) and weights of each of the gaussian components ($\boldsymbol{\omega}_i$) is updated iteratively till the likelihood for the data points (i.e. feature vectors) is maximized, thereby creaing a trained model with the extracted features. The maximized likelihood will thus be assigned to the class corresponding to the speaker, in turn creating the the classified GMM for the speaker.

2.2 Feature Extraction in SR: Mel-Frequency Cepstrum Coefficients(MFCC)

The features which are extracted from the voice samples, are called, *MFCC(Mel-Frequency Cepstrum Coefficients).* MFCCs have become a prominent feature to be used for feature extraction from train data, because it gives an idea about the perceived difference between frequencies, especially in higher frequency region. Its importance comes from the fact that, *the human ear can perceive voice signal frequncies non-linearly.* From here, there is a clear understanding that while using Signal-Processing in this case, frequencies needs to be analyzed in the *Mel* scale, rather than the Hertz scale. There is logarithmic(non-linear) relationship

between the Mel and Hertz scale, from which frequncies in Mel scale becomes more useful. The extraction of MFCCs can be theoretically elaborated in the steps below:

- **Pre-Emphasis:** At first, Pre-Emphasis(after discretization of the analog voice signal) is done to increase the energy of the signal at higher frequencies, the frequency range where the human ear tends to perceive less in linear scale. Pre-Emphasis is mainly done for transmission cases which brings the significance of prescence of higher frequency components in the voice signal. This process is done by passing the signal through a *First-Order High Pass Filter*. The equation for the signal (x[n]) and its output signal (y[n]) after passing through the filter is below:

$$\boldsymbol{y[n] = x[n] - \alpha x[n-1]} \, \forall \, \boldsymbol{\alpha} \in (0.9, 1) \tag{4}$$

  Thus, the z-transform of the filter will be :

$$\boldsymbol{H(z)} = 1 - \boldsymbol{\alpha z}^{-1} \tag{5}$$

- **Framing:** Due to changes in *Prosody*(features of voice) and *random variations* in the vocal tract, the voice signal is a *non-stationary signal(or, random signal)*. However, within short intervals the voice signal is assumed to be *stationary*, for which this signal is analysed over short time windows. Hence, for analysis of voice features, framing is required, as it divides the whole speech signal into short frames of **N** samples and the adjacent frames are overlapped by **M** samples($N > M$).

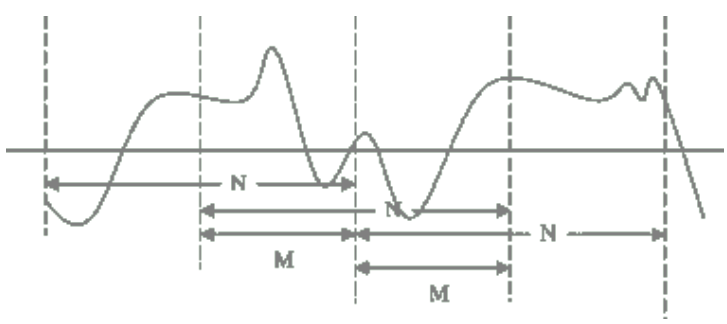

Figure 1: Framing of Voice signal

- **Windowing:** This process is done to remove discontinuities between frames of the audio signal. When the whole signal is divided into short frames, discontinuities are formed at the edges of the frame, which are incongruent to the input voice signal. These discontinuities may impact the signal by changing the **Statistical Properties** of the signal. In order to avoid this problem, windowing is executed by multiplying each frame with a window, a smooth function which goes zero at the borders. Generally,

the signal frames are multiplied with the *Hamming Window*. Considering the window to be w[n], the equation for windowing is :

$$y[n] = x[n] \cdot w[n] \tag{6}$$

where, $\boldsymbol{x[n]}$ is the Input Signal, $\boldsymbol{y[n]}$ is the Output Signal and for N being the number of samples in each frame,

$$\boldsymbol{w[n]} = 0.54 - 0.46 \cdot \mathbf{cos}\,\frac{\boldsymbol{2\pi n}}{\boldsymbol{N-1}}\,\forall\,\boldsymbol{n}\,\in\,[\mathbf{0}, \boldsymbol{N-1}] \tag{7}$$

- **Performing the FFT(Fast Fourier Transform):**We need to execute the *FFT or N_1-point DFT(Discrete Fourier Transform)* of each frame of the signal to obtain their frequency spectrum from the absolute value of the DFT(time domain to frequency domain). N_1 should be greater than the total sample size of each frame. The spectrum plot is considered from k = 0,.....,$\frac{N_1}{2}$, where $f_{Hz} = k\frac{f_s}{N_1}$, f_s being the sampling frequency and $k\frac{f_s}{N_1}$ is called Resolution. The formula for the FFT can be expressed as:

$$\boldsymbol{X[k]} = \sum_{n=0}^{N_1-1} \boldsymbol{x[n]} \cdot \mathbf{exp}\,\frac{\boldsymbol{-j2\pi nk}}{\boldsymbol{N_1}}\,\forall\,\boldsymbol{k} = 0, 1, ..., \frac{\boldsymbol{N_1}}{\mathbf{2}} \tag{8}$$

where, $\boldsymbol{x[n]}$ is the input signal(or, short frame) and $\boldsymbol{X[k]}$ is the FFT of the input x[n]. Here, $|\boldsymbol{X[k]}|$ gives the frequency spectrum of the frames of the input voice signal.

- **Mel-Filter Bank Processing:** The frequency spectrum is then passed through a Mel-filter bank, which is a bank of triangular filters plotted against frequencies in Mel Scale. The relationship between frequencies in Hz scale and frequencies in Mel-scale can be expressed as:

$$\boldsymbol{f}_{Mel} = 2595 \log\,(1\,+\,\frac{\boldsymbol{f}_{Hz}}{700}) \tag{9}$$

The filter output we get after passing the frequency spectrum into Mel-filter bank gives the Mel-Frequency Spectrum. The equation for the filter output $\tilde{S}(l)$ is given below:

$$\boldsymbol{\tilde{S}(l)} = \sum_{k=0}^{N_1/2} \boldsymbol{S(k)} \cdot \boldsymbol{M_l(k)}\,\forall\,\boldsymbol{l} = 0, 1, ..., \boldsymbol{L} - 1 \tag{10}$$

where, $\boldsymbol{S(k) = |X[k]|}$ is the Frequency Spectrum for the input $\boldsymbol{x[n]}$ and $\boldsymbol{L}$ is the number of Triangular Mel-filters used for filtering the whole spectrum to obtain the Mel-frequency spectrum over the whole range of k. The spectrum obtained from filtering operation renders non-linear frequency spectrum at the higher frequencies.The Mel-filter bank usage is important as Voice signal generally follows frequencies in non-linear form and thus the human ear can perceive frequencies being non-linear.

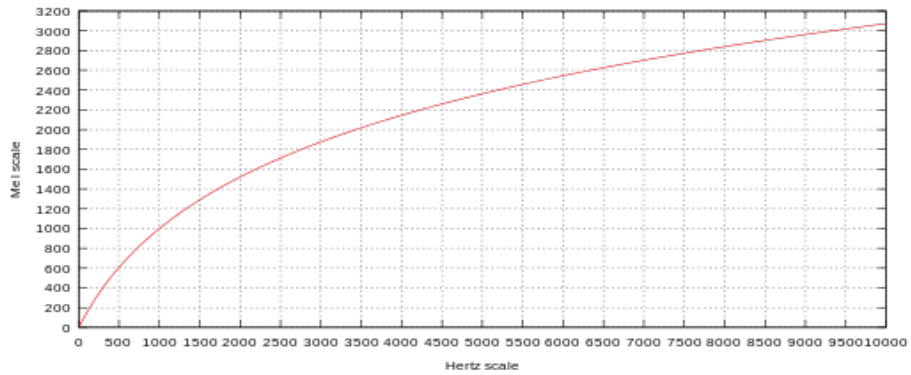

Figure 2: Relationship between Mel-scale and Hz-scale

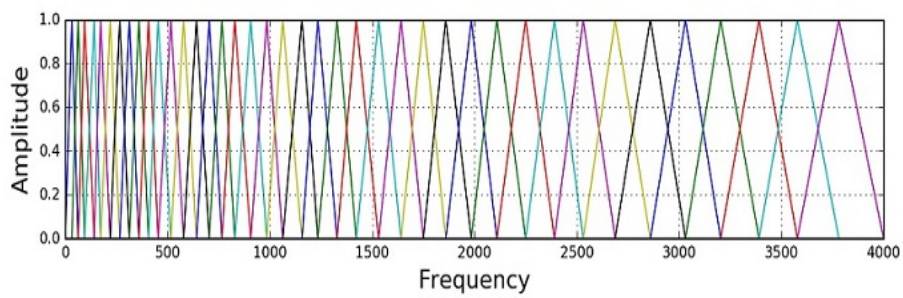

Figure 3: Triangular Mel-Filter Bank

- **Conversion of Mel- Spectrum to Mel-Cepstrum:** The logarithm of the Mel-frequency spectrum is taken to obtain Mel-Frequency Cepstrum.

Figure 4: Conversion of Spectrum to Cepstrum

- **Obtaining the MFCCs:** The Cepstrum obtained in previous step, is operated upon by taking its *Discrete Cosine Transform(DCT)* which provides us MFCCs as the final result. The equation for the performed DCT is as follows:

$$\boldsymbol{c(i) = \sqrt{\frac{2}{L}} \sum_{m=1}^{L} \log\left(\tilde{S}(m)\right) \cdot \cos\left(\frac{\pi i}{L}(m - 0.5)\right) \forall\, c = 0, 1, ..., C - 1} \tag{11}$$

where, $\boldsymbol{C}$ is number of MFCCs desired, $\mathbf{log}\,(\tilde{\boldsymbol{S}}(\boldsymbol{m}))$ is the Mel-Cepstrum obtained and $\boldsymbol{L}$ is the number of triangular mel-filters used for obtaining the Mel-Spectrum.

3 Results and Inference

In the present work, GMM based Speaker Recognition is tested for its efficacy over 10 speakers. The entire coding is done in a Python environment. It is observed that GMM based SR is highly accurate (100% approx).Hence, it can be inferred that this approach for accomplishing Speaker Recognition(SR) is highly efficient and can be used in practical applications where SR is important.

4 Conclusion on Work Done

Speech Recognition is a *text-dependent* method, where it highly depends on the language and corpus. On the other hand, Speaker Recognition mainly focusses on raw audio percepts (and information derived therein) to identify the uniqueness aspect, if any, among different speakers. *Machine Learning* is a holistic approach which renders *Speaker* Recognition efficient when compared to *Speech* Recognition methods for Speaker idenfitication applications. In this work, GMM based ML is applied for classifying the active speaker identification problem. It is found that 100% accuracy may be achieved when tested over 10 different speakers.